

JULIA GONTIJO LOPES

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EDUCATION

New York University -

- Master of Science (MScS), Computer Science (GPA 3.55)

New York, NY

December 2026

Pontifical Catholic University of Minas Gerais

- Bachelor of Science (BS), Computer Science (GPA 3.58)
- Teaching assistant - Algorithm and Data Structure III

Belo Horizonte, Brazil

August 2024

July 2021 - December 2021

Hanyang University of South Korea

- Student exchange program. (GPA 3.79)

Ansan, South Korea

July 2022 - December 2022

WORK EXPERIENCE

Brahma AI - Machine Learning Research Intern

USA | June 2026 - August 2026

- Contribute to Brahma AI's production multimodal data-extraction pipeline for training human-clone video-generation models.
- Profiled and optimized a production multimodal video-ingestion pipeline running on multi-GPU infrastructure, reducing processing time by 56%, from 28 to 12 minutes, and increasing throughput 2.5× across 5,122 clips and 816K frames.
- Optimized landmark storage costs by studying reduced-precision representations, validating their impact on rendering fidelity.
- Owned the evaluation and integration of an AHuBERT-based audio-visual synchronization metric into the production data pipeline, automatically detecting and filtering dubbed and off-sync videos to improve training-data quality for human-video generation models
- Built an end-to-end integration test and containerized it with Docker for isolated CI/CD execution, reducing deployment risk and improving release reliability, and contributed to code reviews across the production codebase.

PUC Minas - Machine Learning Researcher

Brazil | August 2023 - June 2024

- Improved CSSE, a genetic algorithm that generates counterfactual explanations in machine learning interpretability.
- Developed a density-based ranking algorithm for counterfactual generation, prioritizing high-density candidates as more plausible.
- Integrated path-density ranking through nearby real instances, translating feature changes into actionable recommendations.

Bwtech - Software Engineering Internship

Brazil | April 2022 - July 2022 and February 2023 - September 2023

- Contributed to back-end API development for Netchart PM, a platform for monitoring mobile network operators' performance
- Introduced JSON-configurable filtering parameters across data selection, transformation, and KPI-calculation workflows.
- Optimized high-volume telecom-data processing to generate dashboards quickly while preserving API consistency and data accuracy.

Neurau (Early-Stage AI Startup) - Software Engineering Internship

Brazil | May 2024 - September 2024

- Led customer discovery for a ChatGPT-powered WhatsApp chatbot through interviews, surveys, and workflow observation to identify onboarding friction and unmet user needs.
- Applied customer insights to redesign onboarding UX and built an intuitive UI for chatbot personality configuration.
- Provided technical support to the startup's first B2B customers, troubleshooting setup and configuration issues.

Maju Beachwear - Founder

Brazil | October 2020- December 2021

- Founded and scaled an online woman's clothing brand. Profit > \$20K ARP.
- Negotiated with vendors, managed inventory and finances, researched the beachwear market and oversaw products quality control.
- Directed creative marketing campaigns, influencers partnerships, social media branding and photoshoots.

PROJECTS

Visual Attention Modeling for AR/VR Edge Deployment

September 2025 - May 2026

- Used real-time eye gaze for region-of-interest selection in egocentric Meta Quest images for on-device VLM Q&A.
- Reduced downstream tokens by 88.2% and GPU memory usage by 70.6% while accelerating visual encoding.
- Presented the project at NYU's 2026 Research Excellence Exhibition.

Vision-Language Model Efficiency via Token Pruning

September 2025 - December 2025

- Designed a plug-and-play visual-token pruning method for vision-language models using latency and FLOPs profiling.
- Combined text-guided selection of task-relevant regions on image with attention-based pruning of redundant visual tokens
- Reduced latency by 20% and FLOPs by up to 65% while preserving output quality.

Model compression and Knowledge Distillation on Image Classification Models

August 2024 - December 2024

- Trained and fine-tuned VGG16 for image classification on the CIFAR-10 dataset, achieving 88.91% baseline test accuracy.
- Benchmarked 15 structured and unstructured pruning configurations using L1, L2, and random selection at 10-30% pruning.
- Used knowledge distillation to restore or exceed baseline accuracy at up to 30% pruning while keeping inference latency within 1.2%.

Evaluating model-based testing for RL policies

February 2025 - July 2025

- Replicated and evaluated the method proposed in "Test where decisions matter" (Pranger et al., 2024), a model-based testing framework for Reinforcement Learning policies focusing on states where agent's decisions influence on safety outcomes.

Agentic Video Editing Tools

July 2026

- Engineered an MCP server and multi-stage agentic workflow that analyzes multimodal footage, detects scene and action boundaries, translates natural-language direction into frame-accurate edit plans, and automatically renders and validates final videos.

HONORS

Standout Student National Distinction by SBC (Brazilian Computing Society)

August, 2024

Academic prominence distinction in Computer Science (PUC Minas)

August, 2020

SKILLS

Skills: Python, R, Java, C/C++/C#, Scala, SQL, JavaScript, HTML/CSS, .NET, PyTorch, TensorFlow, TorchVision, OpenCV, ONNX Runtime, Apache Spark, Ray, Parallel Computing, Linux/Unix, NVIDIA Nsight Systems, cProfile, Pandas, Git, Docker, Hydra, Windows, Distillation, Scikit-learn, Unity, VLM, AI Agents, Tool Calling, MCP, Agentic Workflows, LLM Orchestration, Multimodal AI, ML Infrastructure, AWS

Languages: Portuguese (fluent)